

P1000 - 1-5/8" x 1-5/8", 12 Gauge Channel, Solid

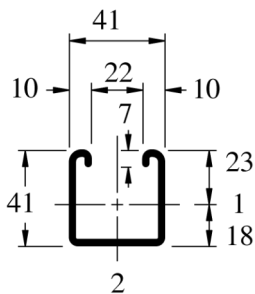
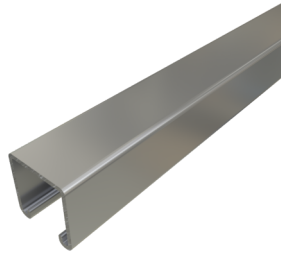
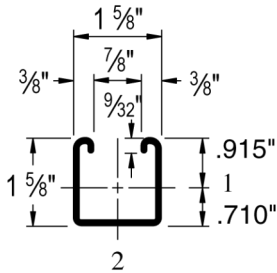
Unistrut's P1000 is the original metal framing strut channel and has been used in countless applications for nearly 100 years. Commonly known as 12 Gauge Standard or Deep Strut Channel, it is the global standard for strut metal framing. This channel is commonly used for trapeze supports, seismic bracing, ceiling grids, pipe, conduit, duct and cable tray supports, racks, and other general framing. For application examples, refer to our Unistrut Application Showcase.

Features

- OPM pre-approved for seismic applications
- UL and CSA listed for use as a Strut-Type Channel Raceway
- Punched holes are also available for ease of installation
- Available in Pre-Galvanized (PG), Atkore Defender (DF), Hot-Dip Galvanized (HG), Plain (PL), Green (GR), Gold (ZD), Stainless Steel (SS or ST) and Aluminum (EA)
- Made in the USA

Applicable Third Party Standards

ASTM



Standard Lengths:

- **10 feet:** 10' or 10' /" (3.05m) ± /" (3 mm)
- **20 feet:** 20' or 20' /" (6.11m) ± /" (3 mm)

Special Lengths:

- Available with a tolerance of ± /" (3 mm). Request quote.

Curved Channel:

- Many Unistrut channel sections can be supplied with a curve. Click here for our ordering form, specifications, and instructions.

Load Data:

- All beam and column load data pertains to carbon steel and stainless steel channels.
- Load tables apply only to UNISTRUT brand channel. Look for "UNISTRUT" on the product.
- Load tables and charts are constructed to be in accordance with the SPECIFICATION FOR THE DESIGN OF COLD-FORMED STEEL STRUCTURAL MEMBERS 2007 EDITION published by the AMERICAN IRON AND STEEL INSTITUTE USING ASD METHOD.
- Loads are based on 33 ksi steel cold formed to 42 ksi.
- Safety Factor to Yield Strength is 1.67 for Beam Loads and 1.80 for Column Loads.
- Beam loads are based on a simple beam and are given as a total uniform load (W) in pounds. For proper calculation procedures, refer to our Beam Load Calculation Guide under Resources.
- For bearing loads, reference our Bearing Loads Page.

Materials & Finishes - Standard:

- **Pregalvanized (PG):** Conforms to ASTM A653 SS GR 33, G90.
- **Unistrut Defender (DF):** Conforms to ASTM A1046 SS GR 33
- **Hot Dip Galvanized (HG):** Steel conforms to ASTM A1011 SS GR 33, Finish conforms to ASTM A123
- **Perma-Green (GR):** Steel conforms to ASTM A1011 SS GR 33, E-Coat finish
- **Perma-Gold (ZD):** Steel conforms to ASTM A1011 SS GR 33, Finish conforms to ASTM B633, Type II SC3
- **Plain (PL):** Conforms to ASTM A1011 SS GR 33

Materials & Finishes - Special Metals:

- **Stainless Steel, Type 304 (SS):** ASTM A240, Type 304 *
- **Stainless Steel, Type 316 (ST):** ASTM A240, Type 316 *
- **Aluminum (EA):** ASTM B221, Type 6063-T6 (Extruded) *

* These materials have different physical properties and performance characteristics. Please contact us for design support.



Catalog Number	Length (ft)	Gauge	Material Type	Surface Finish	Part Weight (lb/ft)	Standard Package Qty (ft)	Standard Package Weight (lb)
P1000 10DF	10	12	Steel	Defender	1.91	500	955
P1000 10EA	10	12	Aluminum		0.75	500	375
P1000 10GR	10	12	Steel	Green E-Coat	1.91	500	955
P1000 10HG	10	12	Steel	Hot-Dip Galvanized	1.91	500	955
P1000 10PG	10	12	Steel	Pre-Galvanized	1.91	500	955
P1000 10PL	10	12	Steel	Plain/Oil	1.91	500	955
P1000 10SS	10	12	Stainless Steel - 304		1.91	500	955
P1000 10ST	10	12	Stainless Steel - 316		1.9	500	950
P1000 10ZD	10	12	Steel	Zinc Dichromate	1.91	500	955
P1000 20DF	20	12	Steel	Defender	1.91	1000	1910
P1000 20EA	20	12	Aluminum		0.75	1000	750
P1000 20GR	20	12	Steel	Green E-Coat	1.91	1000	1910
P1000 20HG	20	12	Steel	Hot-Dip Galvanized	1.91	1000	1910
P1000 20PG	20	12	Steel	Pre-Galvanized	1.91	1000	1910
P1000 20PL	20	12	Steel	Plain/Oil	1.91	1000	1910
P1000 20SS	20	12	Stainless Steel - 304		1.91	1000	1910
P1000 20ST	20	12	Stainless Steel - 316		1.91	1000	1910
P1000 20ZD	20	12	Steel	Zinc Dichromate	1.91	1000	1910

Beam Loading - P1000						
Span (in)	Max Allow. Uniform Load (lbs)	Deflection at Uniform Load (in)	Uniform Loading at Deflection			Lateral Bracing Reduction Factor
			Span/180 (lbs)	Span/240 (lbs)	Span/360 (lbs)	
24	1,690	0.06	1,690	1,690	1,690	1.00
36	1,130	0.13	1,130	1,130	900	0.94
48	850	0.22	850	760	500	0.88
60	680	0.35	650	480	320	0.82
72	560	0.5	450	340	220	0.78
84	480	0.68	330	250	160	0.75
96	420	0.89	250	190	130	0.71
108	380	1.14	200	150	100	0.69
120	340	1.40	160	120	80	0.66
144	280	2.00	110	80	60	0.61
168	240	2.72	80	60	40	0.55
192	210	3.55	60	50	NR	0.51
216	190	4.58	50	40	NR	0.47
240	170	5.62	40	NR	NR	0.44
Note	NR - Not Recommended					

Refer to the General Specifications for loading information.

Column Loading - P1000					
Unbraced Height (in)	Allowable Load at Slot Face (lbs)	Max Column Load Applied at C.G.			
		K=0.65 (lbs)	K=0.80 (lbs)	K=1.0 (lbs)	K=1.2 (lbs)
24	3,550	10,740	9,890	8,770	7,740
36	3,190	8,910	7,740	6,390	5,310
48	2,770	7,260	6,010	4,690	3,800
60	2,380	5,910	4,690	3,630	2,960
72	2,080	4,840	3,800	2,960	2,400
84	1,860	4,040	3,200	2,480	1,980
96	1,670	3,480	2,750	2,110	1,660
108	1,510	3,050	2,400	1,810	KL/r>200
120	1,380	2,700	2,110	KL/r>200	KL/r>200
144	1,150	2,180	1,660	KL/r>200	KL/r>200

Refer to the General Specifications for loading information.

Elements of Section - P1000		
Area of Section	0.555 in ² (3.6 cm ²)	
	Axis 1-1	Axis 2-2
Moment of Inertia (I)	0.185 in ⁴ (7.7 cm ⁴)	0.236 in ⁴ (9.8 cm ⁴)
Section Modulus (S)	0.202 in ³ (3.3 cm ³)	0.290 in ³ (4.8 cm ³)
Radius of Gyration (r)	0.577 in (1.5 cm)	0.651 in (1.7 cm)